

Lecture 8

Attendance Questions

- Tip: Have a mental list of functions with various properties like functions that are continuous but not uniformly continuous (x^2), but also the same conditions but bounded ($\sin(x^2)$)
- Diameter of a subset $K \subset M$ is the $\text{LUB}(\{d(x, y) : x, y \in K\})$

Min-Max Theorem / EVT

- Metric space M is compact and nonempty, $f : M \rightarrow \mathbb{R}$ is continuous
- $f(M)$ is compact, and any compact subset is closed and bounded
- Contains LUB and GLB, which are the maximum and minimum values, respectively
- Think of mapping M to $\mathbb{R}$ as a number line, the GLB $f(M)$ and LUB $f(M)$ are included in the interval

Nested Sequences

- Sequence of nonempty subsets of metric space M
 - Technically nested downward, but nested upward is not very useful
- $A_1 \supset A_2 \supset A_3 \supset \dots$
- Intersection defined as $A = \bigcap_{n=1}^{\infty} A_n$

Theorem 1: If each A_n is compact and nonempty, then A is compact and nonempty

- Choose $a_n \in A_n$
- (a_n) is a sequence in A_1
- Has some subsequence $(a_{n_k})_{n_k \geq n}$, which is a sequence in all A_n
- For each n , this sequence converges in A_n
- Thus, the same limit point is included in A

Nested Sequences of Open Sets

- Metric space M is complete
- All U_i are open, can have a sequence $U_1 \supset U_2 \supset U_3 \supset \dots$
- Suppose $\text{diam } U_n \rightarrow 0$ as $n \rightarrow \infty$
- Choosing points in each U_n , $\text{diam } U_n \rightarrow 0$ ensures this sequence is Cauchy
- Because M is complete, the Cauchy sequence converges
- General assertions about elements are harder to make here about intersection
 - Intersection of nonempty open sets $U_n = (0, \frac{1}{n})$ in $\mathbb{R}$ has empty intersection

Perfect Metric Spaces

- Nonempty metric space M is perfect if every point is a cluster point
- In other words, $\forall p \in M \forall r > 0 \exists q \neq p (q \in M_r(p))$
- Equivalent to $\exists$ sequence (q_n) of distinct points in M that converge to p

Extension vs. Closure of Neighborhoods

- $\hat{x}$ means extension, $\bar{x}$ means closure (x is used loosely here)
- $\widehat{M}_r(p)$ is the closed neighborhood of point p with radius r , whereas $\overline{M_r(p)}$ is the closure of the open neighborhood of point p with radius r
- $\overline{M_r(p)} \subset \widehat{M}_r(p)$
- Think of the set formed by the closed unit disc and the circle with radius 2.
 - Let p be the origin and let r be 2.

- $M_r(p)$ = the open unit disc
- $\overline{M_r(p)}$ = the closed unit disc
- $\widehat{M_r(p)}$ = the closed unit disc and the circle of radius 2

Theorem 2: If a metric space M is perfect & complete, then it is uncountable

- Suppose not.
- $\exists$ distinct $x_1, x_2, x_3, \dots$ s.t. $\{x_1, x_2, x_3, \dots\} = M$
- Look at closed neighborhoods, each excluding a single point of M
- Create a nested sequence $N_1 \supset N_2 \supset \dots$
- Each N_i excludes x_i
- Thus, N_n excludes x_1 to x_{n-1}
- Center points of N_i form a Cauchy sequence, so it converges to some point due to completeness
- Process should exclude all points, but we just showed there is one so we have a contradiction

Connectedness

- A set $S \subset M$ is clopen if it is open and closed in M
- It is also proper if it's nonempty
- S is connected if $\nexists$ proper clopen subset
- S is not connect if $\exists$ proper clopen subsets A, B s.t. $A \sqcup B$
 - $\sqcup$ denotes that A, B are disjoint
- Look at $f : M^{\text{connected}} \xrightarrow{\text{continuous}} N$
 - If f is homeo, then N is connected, as connectedness is invariant over homeo
 - Surjection is sufficient for N to be connected
 - Proof by contradiction
 - Suppose N is disconnected
 - $N = A \sqcup B$
 - We can look at the pre-images $f^{\text{pre}}(A)$ and $f^{\text{pre}}(B)$
 - $M = f^{\text{pre}}(A) \sqcup f^{\text{pre}}(B)$
- Assume N is compact and disconnected
 - Has at least two connected parts A, B
 - $N_r(A)$ is the union of all r -neighborhoods of points in A
 - r is small enough such that $N_r(B) \cap N_r(A) = \emptyset$
 - Not sure what the assertion he made here was
- Corollary of above: $N = \mathbb{R} \Rightarrow \text{IVT}$
 - M is connected & f is continuous
 - For $a, b \in M$, $f(a) < c < f(b)$
 - Normally, $\exists \theta \in M$ s.t. $f(\theta) = c$.
 - If not, then $A = f^{\text{pre}}(f(a), c)$, $B = f^{\text{pre}}(c, f(b))$
 - Then, the real line is disconnected.

Coverings

- Topological approach to connectedness
- $A \subset M$, M is a metric space
- A family of subsets $\mathcal{U}$ of M covers A if A is contained in their union
- Can say that $\mathcal{U}$ is a covering of A
- If it is a family of open subsets, then it is an open covering
- For a subset $\mathcal{V} \subset \mathcal{U}$, if $\mathcal{V}$ also covers A then $\mathcal{V}$ is a subcovering from $\mathcal{U}$

- Language like $\mathcal{V}$ is a subcovering of $\mathcal{U}$ or $\mathcal{U}$ has a subcovering of $\mathcal{V}$

Theorem 3: $A \subset M$ is compact w.r.t sequence iff every open covering of A reduces to a finite subcovering

- Just looking at $\Leftarrow$ (covered in office hours, not lecture)
- Suppose not.
- $\exists$ sequence x_n of A that does not have a convergent subsequence
- $\forall n \exists r_n$ s.t. $M_{r_n}(x_n)$ only contains x_n is a covering of the sequence
- By covering compactness, this sequence can be condensed to a finite subset $\{M_{r_1}(x_1), M_{r_2}(x_2), \dots, M_{r_m}(x_m)\}$
- This statement contradicts the initial condition is wrong, as each r_n -neighborhood is defined to only have x_n
- By the pigeonhole principle, at least one of the m neighborhoods contains at least 2 points
- Thus, x_n must have a convergent subsequence